Fiscal Year:	FY 2007	**Task Last Updated:**	FY 05/17/2007
PI Name:	Halloran, Bernard Ph.D.		
Project Title:	The Mouse as an Animal Model for Human Bone Loss During Spaceflight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94121	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	08/01/2004	**End Date:**	07/31/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE through 7/31/2008 per A. Chu (4/25/2007) No-cost extension until 7/31/2007, per PI (8/14/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CK55G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Spaceflight (skeletal unloading) is associated with loss of bone mass. The mechanisms for this loss are not known. Our objective is to develope an animal model that mimics human bone loss during spaceflight. The goal is to use such a model to study how unloading affects bone cell function and mass. We have chosen to work with the mouse because it is small and in other studies has ben shown to mimic the human with respect to aging, and responsiveness to various endocrine factors. Because there are numerous mouse strains and each behaves differently, we will evaluate three strains. we will use the hindlimb unloading model to induce skeletal unloading and follow the mice over time. Bone formation, mass and structure will be assessed. To determine whether the mouse is a suitable model to study adult human bone loss during spaceflight and recovery after re-ambulation. It is essential to have an animal model with which to study the loss of bone that occurs with spaceflight. Without this it is not possible to determine the mechanisms and thus devise means of reducing bone loss and restoring bone after flight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Disuse osteoporosis is associated with sedentary lifestyles as well as spaceflight and contibutes to fractures of the hip and spine. Our studies are directly applicable to disuse osteoporosis. Understanding how bone responds to loading through animal models will provide a means of defining mechanisms.
Task Progress:	Our goal was to determine whether the mouse is a suitable model to study adult human bone loss during spaceflight, and to examine the effects of reloading in animals treated with IGF-I. MicroCT analysis, bone histomorphometry, and cell culture were used to assess bone structure and makeup, bone formation and osteoprogenitor number and osteoclastogenesis. DBA mice had a lower bone volume than C57 mice. With unloading both lost bone proportionately. Bone formation decreased but did not reach significance. Bone mineral density both total and segmented decreased in the C57 but not in the DBA. Osteoprogenitor number decreased initially and then gradually returned to the control level in both strains. Osteoclastogenesis increased to the same degree in both strains and then gradually returned to normal. Our data indicate that both strains respond to unloading but with important differences. In both strains osteoblast formation decreases and osteoclast formation increases. These changes are consistent with the trend for bone formation to decrease with time of unloading. That bone mineral density decreased in the C57 but not in the DBA suggests that the response of these two strains are not identical. We also examined the effects of reloading after a period of unloading in animals treated with IGF-I. Our previous studies have shown that reloading increases bone formation and gradually restores bone mass. The process is, however, slow and complete restoration of bone may never occur. To determine if we could increase bone recovery and reduce fracture risk, we treated animals with IGF-I during the recovery period. Our findings, which are now in press (see bibliography), suggest that IGF-I treatment speeds bone recovery by stimulating bone formation. Thus IGF-I administration may be of benefit in restoring bone in astronauts after long duration space missions, and in treating patients following a period of immobilization when such patients are returned to normal weight bearing.
Bibliography Type:	Description: (Last Updated: 02/25/2014)
Articles in Peer-reviewed Journals	Boudignon BM, Bikle DD, Kurimoto P, Elalieh H, Nishida S, Wang Y, Burghardt A, Majumdar S, Orwoll BE, Rosen C, Halloran BP. "Insulin-like growth factor I stimulates recovery of bone lost after a period of skeletal unloading" J Appl Physiol. 2007 Apr 5; [Epub ahead of print] PMID: 17412794 , Jun-2007
Articles in Peer-reviewed Journals	Cao JJ, Kurimoto P, Boudignon B, Rosen C, Lima F, Halloran BP. "Aging impairs IGF-I receptor activation and induces skeletal resistance to IGF-1." J Bone Miner Res. 2007 May 8; [Epub ahead of print] PMID: 17488198 , May-2007